Software Development — Guided Exercise 2

Cross-Correction: Method 1 EC/BV Sheet

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1. Overview

This document presents our cross-correction of the Method 1 EC/BV sheet submitted by the assigned partner group. The review is based on the functional requirements defined in the GE2 specification for the register_project method, which covers the six input parameters (company_cif, project_acronym, project_description, department, date, and budget) and the method's outputs.

2. General Assessment

The submitted sheet is largely complete and well-structured. All six input parameters are covered, output classes are included, and the formatting is consistent throughout. The use of interior boundary values for acronym and description length reflects a thorough approach. The main area for improvement is the granularity of the CIF validation rules.

3. Detailed Feedback

3.1 company_cif

- Positive: Data type, length, CIF algorithm, and boundary values are all covered.
- Observation: The sheet separates CIF Valid and CIF Format Valid into two rows. This distinction is somewhat ambiguous and could be clarified or merged into one rule.
- Missing: No dedicated equivalence classes for the CIF sub-components — specifically (1) first character must be a letter A-Z, (2) positions 2-8 must all be digits, and (3) the last character must pass the control check. These are distinct failure modes that warrant separate classes.

3.2 project_acronym

- Positive: Length range 5-10 is correctly covered with lower and upper boundary values.
- Positive: Valid character set (A-Z, 0-9) and its invalid counterpart are correctly stated.
- Observation: Interior boundary values (6, 9 chars) go beyond what is strictly required but are not incorrect.

3.3 project_description

- Positive: Data type, length range 10-30, and boundary values are all present and correct.
- Observation: Same observation as the acronym regarding interior boundary values.

3.4 department

- Positive: All four valid values (HR, FINANCE, LEGAL, LOGISTICS) are listed as separate valid classes. This is correct and well-structured.
- Positive: Data type check is present.

3.5 date

- Positive: DD, MM, and YYYY ranges are all covered with valid ECs, invalid ECs, and boundary values on both sides.
- Positive: The valid date, according to the request date constraint, is correctly identified.
- Observation: The YYYY upper bound is stated as YYYY < 2028, which is equivalent to YYYY <= 2027. Both are correct per the spec.

- Observation: The invalid format class only lists MM/DD/YYYY as an example. It would be more complete to acknowledge other possible wrong formats, such as YYYY-MM-DD or DD-MM-YYYY.

3.6 budget

- Positive: Data type, decimal places, value range, and boundary values are all correctly covered.
- Positive: Splitting less than 2 decimals and more than 2 decimals into separate invalid classes is a more precise approach than a single class.

3.7 Output

- Positive: Output value (MD5 string) and output file (JSON) are both covered as valid classes.
- Positive: All six specific exception messages for invalid inputs are listed, and the duplicate project exception is correctly identified.

4. Summary

The submission is well-executed overall. The primary gap is in the CIF section, where the algorithm check should be decomposed into its sub-components to yield more precise and individually testable equivalence classes. All other parameters are complete or exceed the minimum required coverage.

5. Other Groups Image/Sheet

Data type	RULE	VALID CLASSES	INVALID CLASSES
Company CIF	Data type	ECV1. String	ECNV 1. Different From String
Company CIF	CIF Valid	ECV2. CIF Valid Algorithm	ECNV 2. CIF Not Valid Algorithm
Company CIF	Length is 9	ECV3. Length = 9 BVV1. 9 char	ECNV3a. Length > 9 ECNV3b. Length < 9 BVNV1. 10 Char BVNV2. 8 Char
Company CIF	CIF Format Valid	ECV4. CIF Format Valid	ECNV4. CIF Format not Valid
Project Acronym	Data type	ECV5. String	ECNV5. Different From String
Project Acronym	Length is minimum 5 and maximum 10	ECV6. 5 <= Length <= 10 BVV2. 5 Char BVV3. 6 Char BVV4. 10 Char BVV5. 9 Char	ECNV6a. Length < 5 ECNV6b. Length > 10 BVNV3. 4 Char BVNV4. 11 Char
Project Acronym	Valid String (0-9A-Z)	ECV7. Valid String	ECNV7. Not Valid String
Project Description	Data type	ECV8. String	ECNV8. Different From String
Project Description	Length is minimum 10 and maximum 30	ECV9. 10 <= Length <= 30 BVV6. 10 Char BVV7. 11 Char BVV8. 30 Char BVV9. 29 Char	ECNV9a. Length < 10 ECNV9b. Length > 30 BVNV5. 9 Char BVNV6. 31 Char
Department	Data type	ECV10. String	ECNV10. Different From String
Department	Valid Content	ECV11a. "HR" ECV11b. "FINANCE" ECV11c. "LEGAL" ECV11d. "LOGISTICS	ECNV11. Some Other Value
Date	Data type	ECV12. String	ECNV12. Different From String
Date	Date Format	ECV13. "DD/MM/YYYY"	ECNV13. "MM/DD/YYYY"
Date	DD Value	ECV14. 01 <= DD <= 31 BVV10. 01 BVV11. 02 BVV12. 31 BVV13. 30	ECNV14a. DD < 01 ECNV14b. DD > 31 BVNV7. 00 BVNV8. 32
Date	MM Value	ECV15. 01 <= MM <= 12 BVV14. 01 BVV15. 02 BVV16. 11 BVV17. 12	ECNV15a. MM < 01 ECNV15b. MM > 12 BVNV9. 00 BVNV10. 13
Date	YYYY Value	ECV16. 2025 <= YYYY < 2028 BVV18. 2025 BVV19. 2026 BVV20. 2027	ECNV16a. YYYY < 2025 ECNV16b. YYYY >= 2028 BVNV11. 2024 BVNV12. 2028
Date	Valid Date According to Request Date	ECV17. Date is equal to or after the date on which the request to register the project was received	ECNV17. Date is before the date on which the request to register the project was received
Budget	Type of Data	ECV18. Float	ECNV18. String
Budget	Number of Decimal Places	ECV19. Exactly 2 Decimals BVV21. 2 Decimals	ECNV19a. Less than 2 Decimals ECNV19b. More than 2 Decimals BVNV13. 1 Decimal BVNV14. 3 Decimals
Budget	Value between 50000.00 and 1000000.00	ECV20. 50000.00 <= Budget <= 1000000.00 BVV22. 50000.00 BVV23. 50000.01 BVV24. 1000000.00 BVV25. 999999.99	ECNV20a. Budget < 50000.00 ECNV20b. Budget > 1000000.00 BVNV15. 49999.99 BVNV16. 1000000.01
Output	Output Value	ECV21. A string in MD5 format corresponding to the input data	ECNV21a. EnterpriseManagementException - "Invalid Company CIF" ECNV21b. EnterpriseManagementException - "Invalid Project Achronym" ECNV21c. EnterpriseManagementException - "Invalid Project Description" ECNV21d. EnterpriseManagementException - "Invalid Department" ECNV21e. EnterpriseManagementException - "Invalid Date" ECNV21f. EnterpriseManagementException - "Invalid Budget"
Output	Output File	ECV22. Data is stored in the indicated JSON file	ECNV22. EnterpriseManagementException - "Project with same name for the same CIF already existed in the data file "